



NASA satellite for farmers

NASA's Soil Moisture Active Passive (SMAP) satellite will be launched to notify farmers about moisture levels on a global scale <http://dgit.in/FarmingSat>

Raptr updated

AMD's Raptr service gets a new update that enables users to record gameplay and the players reactions simultaneously while gaming: http://dgit.in/Raptr_Up **Tomorrow's tech**

with 1 million artificial neurons connected to each other via 256 million synapses in its architecture. Designed using Samsung's 28 nanometer process, it easily handles highly dense on-chip memory with extremely efficient transistors. Since these neuromorphic chips are designed to mimic the brain they use transistors as artificial neurons. And although the brain is no different in design, it does have, on average, between a hundred trillion and a quadrillion neurons with an additional 19 to 23 billion neurons in the cerebral cortex. Which of course is a significant order of magnitude greater than the IBM SyNAPSE. But they're not done building it just yet.

SkyNet 101: Users' Manual

Despite the unfavourable comparison to a human brain the TrueNorth chip is a great advancement for computing. In addition to the artificial neurons and synapses the chip is powered by 4,096 neurosynaptic cores of integrated memory, all of which constitute 5.4 billion transistors making it the biggest chip ever created. This allows for enhanced computational logic, communication protocols and event-driven asynchronous operations, parallel computing and fault-tolerant architecture. Simply put - it adapts and multitasks like the human brain with greater levels of complexity and stability than any chip ever designed.

While computers measure their performance in floating point operations per second (FLOPS), IBM has devised a new metric for TrueNorth - synaptic operations per second (SOPS). The chip is capable of 46 billion SOPS which easily places it in the supercomputer range. The only dif-



The board connected 16 TrueNorth chips seamlessly through tiling to create a single unified network.

ference is that it does this while being the size of a post-stamp and runs on a battery found in your wristwatch. The power consumption for the chip is just 70 milliWatts, trivial compared to the whooping 250Watts required to power the GeForce GTX 780 GPU chip. This performance to energy consumption ratio makes the IBM SyNAPSE possibly the most energy efficient chip in existence.

The human brain itself consumes about 20 to 40Watts of power which is a benchmark IBM intends on achieving by expanding on the chip's configuration. They have already tested the creation of a prototype network with these chips to gain insight into future neurosynaptic supercomputers. The capabilities of this prototype was successfully executed by deploying a 16-chip system network with 16 million artificial neurons and 4 billion programmable synapses. The goal of crossing 100 trillion synapses in the near future is now virtually certain.

The Direction of TrueNorth

Pattern recognition is the first step to intelligence. Specialised operations like experiential recall, image and voice recognition, qualitative detection of input and many other "intelligent" functions performed by machines normally take sophisticated programming algorithms which require an immense amount of power to function. As in the case of Google's facial recognition technology for videos, we know that it takes 16,000 processing chips using 100 kilowatts of power to distinguish between the face of a human and a cat. For us as humans such tasks are menial and effortless but now, with its biologically inspired design, neuromorphic chips can tackle such tasks as easily as a human.

The proof is in TrueNorth's ongoing development at Stanford University where TrueNorth is wired up to a camera atop the university tower observing passing traffic. The TrueNorth based system demonstrated that it could easily distinguish between a person riding a bike and one walking a bike, and recognise the change of states between them. Just as the brain doesn't run on any software, but learns through experience, so does TrueNorth. In the human brain, a reiterating series of synaptic signals along neurons which are repeatedly used solidify - we call this neural-synaptic link building process as learning. In the same way TrueNorth operates by learning to use the complexity of the neural network within its low energy architecture to form successful learning links. For a machine this is a new step in evolution.

Although the nature of neuromorphic technology is easier to understand we



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